

Setup & Startup

What the app is made of, and what happens in the first few seconds

⚠ FOR UNDERSTANDING PURPOSES ONLY

Prasanaa Kartick T — Personal App Project

Flutter / Dart · Android

 Packages

 Android build

 Theme

 Splash

1. What This Part Covers

Before any screen appears, a handful of behind-the-scenes files decide what the app is called, which ready-made tools it borrows, how it is packaged for Android, how it is coloured, and what the user sees first. None of these files hold personal data — they are the app’s “packing list, ID card and front door”.

Part	Plain-English role
Project settings	The packing list: name, version, borrowed tools, bundled pictures / sounds / fonts, launcher-icon recipe
Android build settings	The instructions that turn the code into an installable Android app
Debug helper	A small development-only add-on that helps the developer’s tools find and open the app
App entry point	The “light switch”: the first thing that runs, plus the app-wide theme and lock-on-return watcher
Boot screen	The branded splash shown for a couple of seconds

2. The Packing List

The app is version **1.0.0** (build 1) and is marked private, so it cannot be accidentally published to the public Flutter package store. Rather than building everything from scratch, it borrows well-tested tools from other developers:

Tool (library)	What it gives the app
Saved settings	Remembers small things on the phone — most of the app’s lists, plus switches like sound on / off
Charts	Draws the graphs on every dashboard

Tool (library)	What it gives the app
PDF maker	Builds invoices and the Daily Log receipt as PDF files
Excel maker	Builds multi-tab spreadsheets for exports
File sharing	Opens the phone's normal Share menu (email, WhatsApp, Drive...)
Folder finder	Finds safe places on the phone to save generated files
File picker	Lets you choose a backup file from anywhere on the phone
Encryption and hashing	Locks backup files and scrambles stored lock codes so the plain secret is never kept
Biometric unlock	Asks the phone's own fingerprint / face system to confirm it's really you
Sound player	Plays the short click and "saved" chime effects
Fonts and icons	A library of text fonts and an Apple-style icon set
Archive helper	Used for one small formatting fix inside generated Excel files

Bundled pictures, sounds and fonts

The finished app carries its own logo images (including red and yellow variants used on Home), the boot-screen artwork, sound effects, and a bold serif font used for the "TIME SHEET" heading on invoices. The Android home-screen icon is generated from a single square logo image.

3. Android Packaging

- The app is built for **Android only** (no iPhone version).
- A compatibility layer lets newer date-and-time features work correctly even on older Android phones.
- Every Android helper module is forced to compile to one common standard, which avoids build errors when different plug-ins expect different versions.
- A development-only add-on lets the developer's tools launch the app directly. It also allows internet access purely so the computer can talk to the app during live code updates; it is not part of a normal install.

4. What Happens When the App Opens

1. The phone runs the app's single starting point, which first makes sure Flutter itself is ready.
2. The app shell is created: name, theme, and the Boot Screen as the first page.
3. The Boot Screen shows for about **2.6 seconds**, then fades into Home.
4. Home immediately asks for the App Lock before showing any buttons (see the Home and App Lock documents).

The theme

Two brand colours — a deep charcoal-navy and a fresh green, matching the logo — are defined once. The app uses Google’s newer “Material 3” style with a light off-white background; flat white cards with rounded corners and a thin outline; navy buttons with rounded edges; and a navy top bar with white text. Because everything inherits from this one place, restyling the whole app is a two-colour change.

Lock on return

Always locked when you come back

One app-wide watcher notices when the app goes to the background and returns. If App Lock is set up, the challenge screen appears immediately on top of whatever was open — even a half-filled form — and the back button cannot dismiss it. Section-level checks (such as the quick PIN check on Family & Vault) only ask once per real app open, not on every return.

The Boot Screen

The splash is purely decorative: a near-black background matched to the artwork’s edge colour, the logo fading in over about 0.7 seconds, and a thin progress bar that fills over the full 2.6 seconds. The bar is intentionally “fake” — nothing is really loading — it just reassures the user the app hasn’t frozen. No user data is read here.